

Math 514, F19
Quiz 2

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) According to the weather forecast there is a 40% chance of rain for each of the next five days. The weather on any given day is independent of the weather on the other days. Let X be the number of rainy days for the next five days. Find $E[X]$, $\text{Var}(X)$, and $P(X \leq 2)$.

Let X_i be the random variable such that $X_i = \begin{cases} 1 & \text{rain on the } i\text{-th day} \\ 0 & \text{otherwise} \end{cases}$

clearly $P(X_i = 1) = 40\%$, $P(X_i = 0) = 60\%$ $\hookrightarrow$ Bernoulli random variable with $P = 0.4$
and $X = \sum_{i=1}^5 X_i$ with $E[X_i] = 0.4$, $\text{Var}(X_i) = 0.4 \times (1 - 0.4) = 0.24$
 $\hookrightarrow$ Binomial random variable with $n = 5$, $P = 0.4$

$$E[X] = \sum_{i=1}^5 E[X_i] = 5 \times 0.4 = 2$$

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^5 X_i\right) = \sum_{i=1}^5 \text{Var}(X_i) = 0.24 \times 5 = 1.2$$

$$\begin{aligned} P\{X \leq 2\} &= P\{X = 0\} + P\{X = 1\} + P\{X = 2\} \\ &= \binom{5}{0} 0.6^5 + \binom{5}{1} 0.6^4 \times 0.4 + \binom{5}{2} 0.6^3 \times 0.4^2 \\ &= 0.07776 + 5 \times 0.05184 + 10 \times 0.03456 = 0.68256 \end{aligned}$$

Problem 2. (5 points) If $\text{Cov}(X_i, X_j) = i \cdot j$, find the covariance $\text{Cov}(X_1 + X_2 + X_3, X_2 + X_4)$ and the correlation $\rho(X_1, X_2)$.

$$\begin{aligned} &\text{Cov}(X_1 + X_2 + X_3, X_2 + X_4) \\ &= \text{Cov}(X_1, X_2) + \text{Cov}(X_1, X_4) + \text{Cov}(X_2, X_2) + \text{Cov}(X_2, X_4) \\ &\quad + \text{Cov}(X_3, X_2) + \text{Cov}(X_3, X_4) \\ &= 1 \cdot 2 + 1 \cdot 4 + 2 \cdot 2 + 2 \cdot 4 + 3 \cdot 2 + 3 \cdot 4 \\ &= 36 \end{aligned}$$

$$\begin{aligned} \rho(X_1, X_2) &= \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1) \text{Var}(X_2)}} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Cov}(X_1, X_1) \text{Cov}(X_2, X_2)}} \\ &= \frac{1 \cdot 2}{\sqrt{1^2 \cdot 2^2}} = \frac{2}{\sqrt{4}} = \frac{2}{2} = 1 \end{aligned}$$